

A110 – The Lepton Masses: Yukawa Ladder, Koide, and Circulant

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.1 Purpose: Lepton Masses from One Parameter

[K] The lepton sector is where the theory performs most strongly and is most sharply testable. From the single parameter $\xi = 4/30000$ and a rational assignment per lepton, the three charged lepton masses follow, their ratios, the Koide relation $Q = 2/3$, and – the sharpest point – a prediction of the τ -mass. This document develops both: the *ladder* that generates the masses, and the *circulant* that sharpens the Koide structure.

.2 The Yukawa Ladder

[K] The basic mass formula is

$$m_i = r_i \xi^{p_i} v, \quad v = 246.22 \text{ GeV (Higgs-VEV),}$$

with rational prefactors r_i and rational exponents p_i per lepton:

Lepton	r_i	p_i	m_i (calculated)
Electron	$4/3$	$3/2$	0.505 MeV
Muon	$16/5$	1	105.1 MeV
Tau	$25/9$	$2/3$	1785 MeV

[K] The exponents form a ladder with fixed steps:

$$p_e - p_\mu = \frac{3}{2} - 1 = \frac{1}{2}, \quad p_\mu - p_\tau = 1 - \frac{2}{3} = \frac{1}{3}.$$

These steps $\frac{1}{2}, \frac{1}{3}$ are the core: they generate the $\sqrt{m}$ -structure on which the Koide formula rests, and they are what recurs throughout the following sections.

.3 Origin of the Prefactors: Resonances on the Torus

[K] The prefactors r_i are not freely chosen. Each lepton corresponds to a standing field mode on the torus, characterised by two integer winding numbers (n_θ, n_ϕ) – the number of circuits along each cycle. This is the resonance condition: only modes with integer winding are permitted, as with any standing wave on a ring.

[K] The prefactor follows directly from the winding numbers:

$$r_i = \frac{n_{\phi,i}^2}{n_{\theta,i}}.$$

For the three leptons:

Lepton	n_θ	n_ϕ	$r_i = n_{\phi,i}^2/n_{\theta,i}$
Electron	3	2	4/3
Muon	5	4	16/5
Tau	9	5	25/9

[K] This explains the “structure in the numerators” visible in the r_i : the numerators 4, 16, 25 are the squares $n_\phi^2 = 2^2, 4^2, 5^2$, the denominators 3, 5, 9 are the n_θ . The r_i are simple ratios of two resonance quantum numbers, not fitted values. The masses of the sector thus follow from ξ , the ladder exponents, and a pair of integer winding numbers per lepton.

[K] The same r_i carry a second, equivalent interpretation: they are the consonant intervals of just intonation. In just (5-limit) tuning, consonant intervals are ratios of small integers with prime factors 2, 3, 5 – and that is exactly what the r_i are:

Lepton	r_i	Prime factors	Just interval
Electron	4/3	$2^2/3$	Perfect fourth (498 ct)
Muon	16/5	$2^4/5$	Min. sixth + octave (814 ct)
Tau	25/9	$5^2/3^2$	Square of the major sixth

[K] This is not an analogy but the same fact seen from two sides: a standing mode on a torus cycle *is* a harmonic resonance, like the overtones of a vibrating string. The winding numbers are the order numbers of these resonances; the $r_i = n_\phi^2/n_\theta$ are the corresponding interval ratios.

[K] The three charged leptons are the *simplest* of these resonances – the low 5-limit consonances. The series does not stop there: higher modes are the remaining fermions. The quarks are further overtones of the same torus: the top is the only overtone above the Higgs ground tone, the bottom with $r = 3/2$ – the perfect fifth – the harmonically simplest quark. There is therefore no cutoff at three modes; there is an ascending resonance ladder, and the leptons are its lowest part.

[K] The *ordering* is therefore necessary, not chosen: resonances are ordered by their order number, ascending winding number means ascending mass. In a theory where particles are resonances, “lightest resonance = electron, next = muon, then tau” is not an additional rule but the resonance structure itself.

[K] The tau prefactor $r_\tau = 25/9 = (5/3)^2$ is the square of the major sixth. An earlier iteration carried $r_\tau = 8/3$ (giving $m_\tau \approx 1712$ MeV, -3.6%); the correct value $25/9$ gives $m_\tau \approx 1785$ MeV (Corrected on 30 Sept. 2026: previously ≈ 1783 MeV – $(25/9)\xi^{2/3} \cdot 246.22$ GeV = 1785.0 MeV, as in the table above; 1783 MeV would correspond to $v = 246$ GeV.) and is recorded corpus-wide.

.4 The Three Mass Ratios from ξ

[K] Because the masses are products $r_i \xi^{p_i} v$, the absolute v cancels in every ratio, leaving pure ξ -geometry:

$$\begin{aligned}\frac{m_e}{m_\mu} &= \frac{r_e}{r_\mu} \xi^{p_e - p_\mu} = \frac{4/3}{16/5} \xi^{1/2} = \frac{5}{12} \xi^{1/2} = 0.004811, \\ \frac{m_\mu}{m_\tau} &= \frac{r_\mu}{r_\tau} \xi^{p_\mu - p_\tau} = \frac{16/5}{25/9} \xi^{1/3} = \frac{144}{125} \xi^{1/3} = 0.05885, \\ \frac{m_e}{m_\tau} &= \frac{r_e}{r_\tau} \xi^{p_e - p_\tau} = \frac{12}{25} \xi^{5/6} = 0.000283.\end{aligned}$$

[K] Against experiment (PDG):

Ratio	from ξ	PDG	dev.
m_e/m_μ	0.004811	0.004836	0.51 %
m_μ/m_τ	0.05885	0.05946	1.02 %
m_e/m_τ	0.000283	0.0002876	1.6 %

All three to roughly one percent, from a single parameter and rational numbers.

.5 Koide as Consequence: The Independent Route

[K] The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}$$

is not an independent empirical symmetry but follows from the ladder. Inserting the three calculated masses – *without* using Koide anywhere – gives

$$Q_{\text{ladder}} = 0.66774,$$

which is 0.16 % from $2/3$. The geometric ladder thus predicts the Koide quantity on its own. That the experimental value then hits exactly $2/3$ is an *independent* confirmation of the exponent structure, not its prerequisite.

[B] The reason Q comes out so accurately is ratio-invariance: since only exponent *differences* enter and v cancels, Q depends solely on $\xi^{p_i - p_j}$ and is free of any absolute scale. This is the purest form in which ξ appears.

.6 Three Routes to $2/3$

[B] At exactly $D = 3$ (the topological dimension of the T^4 sector) three different structures meet at the same value $2/3$:

$$1 - \frac{1}{D} = \frac{2}{3}, \quad \frac{2}{D} = \frac{2}{3}, \quad \frac{1}{2} \left(1 + \frac{1}{D} \right) = \frac{2}{3}.$$

The first is the complement step, the second the Koide equilibrium condition between democratic and orthogonal projection components ($A = B$), the third the ladder ratio. A formal deformation $D = 3 + \varepsilon$ separates the three again – they coincide only at integer $D = 3$. This binds $2/3$ to the *integer* topological dimension, not to the fractal $D_f = 3 - \xi$.

.7 The Circulant: Sharpening to the Tau Mass

[K] Alongside the ladder there is a second, complementary description: the three $\sqrt{m_i}$ as discrete Fourier transforms of a Z_3 -circulant. In this representation two structural quantities appear, an amplitude and a phase angle θ , and the corpus finds

$$\theta = \frac{2}{9}.$$

[K] Two conditions then fix m_τ : the Koide condition $Q = 2/3$ and the phase condition $\theta = 2/9$. Both are formally distinct, but at the current measurement precision they coincide (distance $< 0.05 \sigma$), delivering the same m_τ ; the sharp prediction is

$$m_\tau^{\text{pred}} = 1776.968 \text{ MeV}.$$

[K] The bookkeeping of the sector is decisive here. The lepton masses are *not* inputs to the theory: the structural numbers ($\sqrt{2}$ from $Q = 2/3$, $2/9$ as address) are theory-side. To anchor to the measured pole masses, exactly *one* declared reference point is used: the ratio m_μ/m_e . It fixes the operative angle $\theta_{\text{ref}} = 0.2222220471$ and is thereby consumed as a calibration quantity.

[K] The gain of this one anchor choice is the sharp prediction:

$$m_\tau = 1776.9690 \pm 0.0001 \text{ MeV},$$

input-error-free, $+0.91 \sigma$ against PDG 1776.86 ± 0.12 (Belle-II world average decides). This is the series' actual prediction (A210, A220): a numerical value fixed *before* measurement – not a bookkeeping value and not a scale choice like α or G , but a genuinely measurable quantity, because the ratio structure forces the third mass from the two light ones.

[B] The Q -route and the θ -route are two formally distinct conditions, not one. At current measurement precision they coincide (distance $< 0.05 \sigma$), delivering the same m_τ ; they are separated only by a measurement that determines m_τ more sharply than this bound.

.8 The xi-Cascade: Why the Masses Are Most Precise

[B] The lepton sector stands at the pure end of a hierarchy that orders the whole theory:

$$\xi \xrightarrow{\text{ratios}} Q = \frac{2}{3} \xrightarrow{\text{scale}} \alpha \xrightarrow{\text{SI metric}} G.$$

Each step adds a layer of complexity, and precision falls accordingly:

Quantity	requires	precision
Q (Koide)	ratios only	0.0003 %
α	+ energy scale E_0	0.006 %
G	+ Planck length	0.5 %

(Corrected on 30 Sept. 2026: precision of Q previously 0.00003 % – with PDG 2024 ($m_\tau = 1776.93 \text{ MeV}$), $Q_{\text{exp}} = 0.6666645$, relative deviation from $2/3$: $-3.3 \times 10^{-6} = 0.0003 \%$; with $m_\tau = 1776.86 \text{ MeV}$, 0.0009 %.)

[K] This explains the accuracy classes from A220 from *one* principle: the closer a quantity is to pure ratios, the more accurately it follows from ξ ; the more absolute scale it requires, the more bookkeeping enters. The lepton masses are the purest form and therefore the most precise and the only genuine prediction. α and G are the same ξ -geometry, only in increasingly dressed form (A130, A145).

.9 Verification Script

- The Yukawa ladder $m_i = r_i \xi^{P_i} v$ with the three mass ratios from ξ (range check against PDG); the ladder prediction $Q_{\text{ladder}} = 0.6677$ without Koide input; the two-route sharpening to m_τ (prediction, 0.90σ); and the ξ -cascade with falling precision: `python/A_Serie_Skripte/a110_zirkulant.py`

.10 Symbol Glossary

Symbols used throughout the A-Series are explained in A010. Specific to this document:

Symbol	Meaning
C	3×3 -circulant matrix; diagonalisable via DFT
$r = \sqrt{2}$	eigenvalue ratio; condition for $Q = 2/3$
$Q = (\sum m_i)/(\sum \sqrt{m_i})^2$	Koide operator; measured $Q \approx 2/3$
$\omega = e^{2\pi i/3}$	primitive third root of unity

.11 Open Questions

First: the mode ordering is not an open point. The three charged leptons are the simplest torus resonances, and their ordering follows necessarily from the ascending resonance order (above). Higher modes are the remaining fermions; nothing cuts off at three. What is set – and exclusively that – is the *cardinality* three via the order Z_3 (A020). Z_3 is an axiom, not a derivation, and an axiom needs no written derivation to be valid: every theory enters its own framework at some point. Z_3 is motivated by the parsimony of the three-point configuration in the two-dimensional cross-section and by agreement with the generation number – but these reasons do not make the axiom a derivation, and their absence would not be a deficiency. What is to be avoided is only the reversal: reading the three generations as a *consequence* of Z_3 . In fact the observed number three motivates the choice Z_3 , not vice versa.

Second, the percent scatter of the ladder is not an open point but the reason for the reference-point booking. The ladder is a percent-level statement; a ξ back-calculated from it scatters accordingly. That is exactly why the sector anchors the scale not in the ladder but in *one* reference point m_μ/m_e . The circulant with the pair $(\sqrt{2}, 2/9)$ misses m_μ/m_e at pole-mass level by $1.0 \cdot 10^{-5}$ – which at this level excludes reading the pair as *exact*; as a rational address to seven digits it remains valid. The booking is thus resolved, not open.

Third, $\theta = 2/9$ is precisely located, not unmotivated. $2/9$ is the rational address of the reference angle θ_{ref} to seven digits ($|\theta_{\text{ref}} - 2/9| = 1.75 \cdot 10^{-7}$). It lives in the angular phase sector and corresponds to the phase position $\chi = \pi/2$; the radial mass sector remains rigid. What stands as a candidate – not an achieved result – is only the *dynamic selection*: whether a skew-adjoint generator outside the static geometry locks the phase onto $\chi = \pi/2$. FFGFT's own real recursion carries no such generator; whether an external dynamics supplies it is the only remaining open point – the location and address of $2/9$ are fixed.

Fourth, the dependence on m_τ is the falsifiability, not a deficiency. After the reference-point booking, m_μ/m_e is the consumed anchor and $m_\tau = 1776.9690 \pm 0.0001$ the one sharp test quantity. That the sector stands or falls by it is intentional: a measurement placing m_τ clearly away from 1776.97 refutes the exponent structure (A210). That is the purpose of a prediction, not its weakness.

.12 Supersedes

This document subsumes: Doc. 116 (Koide from ξ , Yukawa ladder, ξ -cascade), Doc. 258 (three routes to $2/3$, $D = 3 + \epsilon$), Doc. 259 (cross-terms), Doc. 060 (musical resonance interpretation of prefactors), Doc. 189 (torus derivation of winding numbers), Doc. 190 (register K2 on r_T , R56 on Z_3), Doc. 282 (circulant, HLV bridge), Doc. 291 (dynamic location of $\theta = 2/9$, $\chi = \pi/2$), Doc. 292 (empirical lepton check). Incorporated: P40 (ratios/bridge constants), P42 (one reference point m_μ/m_e).

.13 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.